

Reflective Journal

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Working on this machine learning project helped me understand the process of building and evaluating a model more than I expected. One of the biggest challenges I faced was dealing with the class imbalance in our stroke dataset. Only a small percentage of the patients had experienced a stroke, which made it difficult for the models to identify stroke cases. I learned that accuracy alone does not always show whether a model is performing well. For example, the Random Forest model had the highest accuracy at 91.66%, but its recall was only 13.16%. This showed me that a model can have a high accuracy while still missing many of the cases that are most important to identify.

I also learned how important it is to properly split the data before applying techniques such as SMOTE. I learned that SMOTE should only be applied to the training data while the validation and test data should remain untouched. This helped me understand data leakage and why it can make a model's results look better than they actually are. I also became more comfortable comparing different models and understanding why we need to look at multiple metrics instead of only choosing the model with the highest accuracy. Logistic Regression had an ROC-AUC of 0.825 and a recall of 81.58%, which made it much better at identifying stroke cases than Random Forest. We also compared the Voting and Bayesian ensemble models, which helped me understand how combining models can produce different results depending on the metric being evaluated.

If I were to continue working on this project, I would spend more time tuning the models and experimenting with different classification thresholds to improve recall. Since this is a healthcare-related problem, I think identifying potential stroke cases is especially important because missing a case could have serious consequences. Overall, this project changed the way I think about machine learning. I originally thought the model with the highest accuracy was automatically the best model, but I now understand that the best model depends on the purpose of the project and which errors are most important to avoid. This project also helped me become more comfortable working through different steps of a machine learning project and gave me a better understanding of how the technical decisions I make can affect the final results.